LESSON 2.12

Multiply Decimals

Set B · Version A · Second form · supported practice

MASTERY CHECK

- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can multiply decimals and place the decimal point correctly in the product.

WHY THIS MATTERS The station needs 4.5 meters of heat shielding that costs $12.60 per meter. Commander Diaz needs to calculate the total cost before submitting the supply order. What is the total cost of the shielding?

REMEMBER

Multiply Decimals

Multiplying Decimals To multiply decimals, you multiply the numbers as if there were no decimal points. Then you count the decimal places and put the point back in the answer.

1. Multiply the numbers as if they were whole numbers (ignore decimals during multiplication).
2. Count the total number of decimal places in both factors.
3. Place the decimal point in the product so it has that same total number of decimal places

Words: **Product *(Producto)*** · **Decimal point *(Punto decimal)*** · **Estimate *(Estimar)*** · **Place value *(Valor posicional)*** · **Decimal places *(Cifras decimales)***

PART 1 Practice the skill

show every step

1 **Circle the letter of the best answer. Show how you know.**

What is 3.2 + 4.5?

(A) 7.5 (B) 7.7

(C) 8.7 (D) 77

LINE UP THE PLACE VALUES

1. **Line up** *Points under points.*
2. **Fill gaps** *Write in the zeros.*
3. **Operate** *Right to left.*
4. **Bring the point down** *Straight down.*

2 Circle the letter of the best answer. Show how you know.

What is 14.6 + 3.85?

(A) 17.45
(B) 18.35
(C) 18.45
(D) 18.55

LINE UP THE PLACE VALUES

1. **Line up** *Points under points.*
2. **Fill gaps** *Write in the zeros.*
3. **Operate** *Right to left.*
4. **Bring the point down** *Straight down.*

PART 2 Apply it

real-world problems

3 Circle the letter of the best answer. Show how you know.

What is 20.3 − 7.48?

(A) 12.18
(B) 12.82
(C) 12.92
(D) 13.82

LINE UP THE PLACE VALUES

1. **Line up** *Points under points.*
2. **Fill gaps** *Write in the zeros.*
3. **Operate** *Right to left.*
4. **Bring the point down** *Straight down.*

4 Circle the letter of the best answer. Show how you know.

What is 50 − 23.47?

(A) 26.53
(B) 26.63
(C) 27.47
(D) 27.53

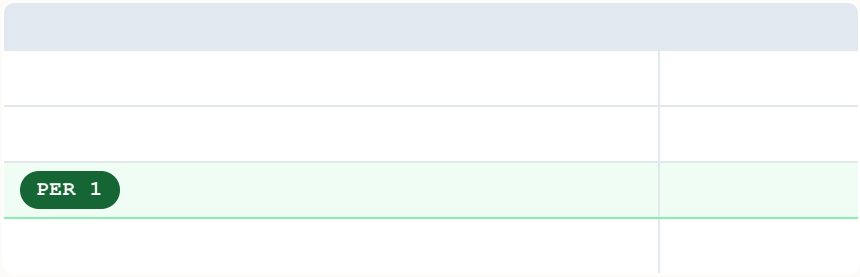
LINE UP THE PLACE VALUES

1. **Line up** *Points under points.*
2. **Fill gaps** *Write in the zeros.*
3. **Operate** *Right to left.*
4. **Bring the point down** *Straight down.*

5 Circle the letter of the best answer. Show how you know.

A family is buying 3.5 pounds of deli meat that costs $6.80 per pound at the grocery store. 3.5 × $6.80 = ?

(A) $2.38
(B) $21.80
(C) $23.80
(D) $238.00



1. **Label** *Write what each column is.*
2. **Find per 1** *Divide to get one.*
3. **Scale** *Multiply up to what is asked.*
4. **Check** *Does the size make sense?*

6 Circle the letter of the best answer. Show how you know.

A family is buying 3.5 pounds of deli meat that costs $6.80 per pound at the grocery store. Round to estimate: about how much should the deli meat cost?

(A) about $2.80
(B) about $14
(C) about $28
(D) about $280

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How did estimating by rounding help you sort the products? How do you decide where to place the decimal in the exact product?

SENTENCE STARTER I estimated ___ x ___ by rounding to ___ x ___ = ___, which told me the product is ___ than 20.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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LEARNING TARGET I can multiply decimals and place the decimal point correctly in the product.

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PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each food truck order to its total.

_____ 4 × $3.75 (combo meals)	(A) $15.00
_____ 5 × $1.80 (sodas)	(B) $9.00
_____ 3 × $2.40 (cookies)	(C) $7.20
_____ 7 × $4.20 (burritos)	(D) $29.40
_____ 0.5 × $0.80 (napkin pack)	(E) $0.40

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

A family is buying 3.5 pounds of deli meat that costs $6.80 per pound at the grocery store. A student answered $238.00. What went wrong?

(A) They added instead of multiplying
(B) They rounded too early
(C) They used the wrong price
(D) They put the decimal point in the wrong place

SHOW YOUR WORK

3 Circle the letter of the best answer. Show how you know.

A customer orders 3 taco baskets at $4.25 each. What is the total before tax?

(A) $1.275
(B) $12.15
(C) $12.75
(D) $127.50

SHOW YOUR WORK

4 Circle the letter of the best answer. Show how you know.

Each lemonade uses 0.6 of a cup of sugar, and sugar costs $0.40 per cup. How much sugar cost is in one lemonade? (0.6 × 0.4)

(A) $0.024
(B) $0.24
(C) $2.40
(D) $24.00

PER 1

5 Circle the letter of the best answer. Show how you know.

The truck sells 2.5 pounds of fries, and each pound earns $1.20 in profit. What is the total profit? (2.5 × 1.2)

(A) $0.30
(B) $3.00
(C) $3.70
(D) $30.00

SHOW EACH PARTIAL PRODUCT, THEN ADD THEM

PART 3 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Level 2 Extension — Find the Cashier's Mistake

1. **Order** 2 burrito bowls at $6.45 each: 2 × 6.45
2. **Multiply as whole numbers** 2 × 645 = 1,290
3. **Count decimal places** 6.45 has 2 decimal places → 2 total
4. **Place the decimal point** 129.0, so the total is $129.00

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How did estimating by rounding help you sort the products? How do you decide where to place the decimal in the exact product?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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